



SLEEPHELPER

PROJECT PROPOSAL & DEVELOPMENT PLAN

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1. Project Overview

SleepHelper is a concept of an interactive hardware-software system designed to support users in developing healthier and more consistent sleep habits.

The project combines an embedded electronic device, environmental sensors and a companion mobile application. The system is intended not only to monitor selected environmental and behavioural factors associated with sleep, but also to actively support the user before bedtime by providing personalised recommendations, reminders and feedback.

The physical device is designed as a bedside or wall-mounted unit incorporating a clock, alarm functionality, environmental monitoring, a dedicated smartphone charging station and a user interface. The accompanying application would collect and analyse data and provide the user with a personalised overview of their sleep habits.

A central element of the project is the concept of **SleepScore** - a non-medical indicator combining selected environmental measurements, behavioural data and the user's subjective assessment of sleep quality.

The project is intended as an experimental engineering and product-design project combining embedded systems, programming, electronics, data analysis, mobile software and industrial design.

2. Problem & Motivation

Sleep is one of the fundamental factors affecting human health, cognitive performance and everyday functioning. Despite its importance, insufficient and poor-quality sleep remains a widespread problem.

The issue cannot be reduced solely to the number of hours spent sleeping. Sleep quality can also be influenced by factors such as regularity of sleep schedule, environmental conditions, exposure to light and behaviours preceding bedtime.

At the same time, modern technology has introduced an additional challenge. Smartphones and other digital devices are frequently used during the evening and are often kept next to the bed. This can make it more difficult to establish a clear boundary between active time and rest.

Many contemporary approaches to improving sleep habits are closely connected to the user's smartphone. While mobile applications provide convenient access to sleep tracking, recommendations and behavioural statistics, they also place the tool intended to support healthier habits on the same device that is often a source of distraction. This creates an inherent conflict: the user may open the phone with the intention of checking their sleep data or following a recommendation, but the same interaction can easily turn into unrelated browsing or entertainment. As a result, it becomes difficult to establish a clear behavioural boundary between using the smartphone as a tool for improving sleep habits and using it purely for entertainment or habitual checking.

Existing sleep-related technologies frequently focus on ***measuring and reporting sleep after it has occurred***. SleepHelper is intended to approach the problem from a slightly different perspective: instead of relying exclusively on a smartphone-based experience or analysing the previous night, it should help the user ***prepare for sleep and develop healthier habits over time through a dedicated physical interface and minimal digital interaction***.

An important element of this approach is therefore ***physical separation from the smartphone***. By incorporating a dedicated charging station into SleepHelper device, the project aims to make it possible for the user to leave their phone outside immediate sleeping environment while still retaining access to essential functions such as an alarm clock and sleep-related information.

The motivation behind the project is therefore to investigate whether a combination of environmental monitoring, behavioural feedback, physical interaction and deliberately limited digital interaction can encourage more consistent sleep-related habits.

3. Project Goal

The primary goal of SleepHelper is to design and develop a functional prototype of an ***interactive system supporting healthy sleep habits***.

The project will explore how an embedded device and mobile application can work together to:

- ◆ Help the user establish a regular sleep schedule,
- ◆ Provide information about recommended sleep times,
- ◆ Monitor selected environmental conditions,
- ◆ Reduce unnecessary smartphone use before bedtime,
- ◆ Provide personalised feedback,
- ◆ Visualize long-term-sleep-related patterns,
- ◆ Encourage consistent bedtime routines.

The project will also investigate how technological functionality can be combined with ***human-centred and aesthetically conscious product design***.

4. Target User

The initial target group consists primarily of people who:

- ◆ Experience irregular sleep schedules,
- ◆ Have difficulty maintaining consistent bedtime routines,
- ◆ Use smartphones extensively during the evening,
- ◆ Want to improve their sleep habits,
- ◆ Are interested in monitoring their environment,
- ◆ Prefer practical technological solutions rather than purely software-based trackers.

The project is not intended to diagnose, treat or replace professional medical assessment of sleep disorders. SleepHelper should instead function as a personal wellbeing and habit-support system.

5. Planned Functionality

5.1 Clock & Time Management

The device will provide:

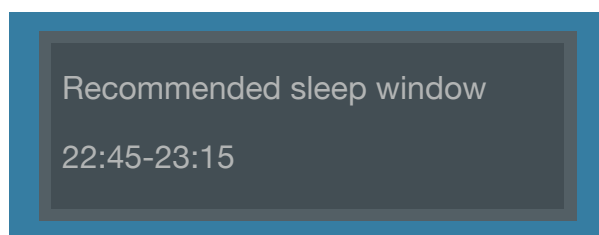
- ◆ Current time,
- ◆ Date,
- ◆ Configurable alarm,
- ◆ Sleep schedule,
- ◆ Recommended bedtime,
- ◆ Wake-up time,
- ◆ Reminders before the planned bedtime.

The clock will serve as the primary interface of the physical device.

5.2 Sleep Window

The system will calculate a recommended sleep window based on the user's desired wake-up time, preferred sleep duration and potentially other user-defined parameters.

For example:



The device may notify the user when their recommended bedtime is approaching.

The exact methodology used to calculate the sleep window will be determined during the research and development phase.

6. Environmental Monitoring

The physical device is planned to monitor selected environments parameters, including:

- ◆ Temperature,
- ◆ Relative humidity,
- ◆ Ambient light.

Future versions may additionally investigate:

- ◆ CO₂ concentration,
- ◆ Ambient noise,
- ◆ Air quality,
- ◆ Movement.

The collected measurements could be used both for real-time information and for longer-term analysis.

For example, the application could identify relationships between environmental conditions and the user's subjective sleep quality.

7. Smartphone Integration

One of the central concepts of SleepHelper is to create a physical separation between the user and their smartphone during sleep.

The device will therefore incorporate a smartphone charging station.

Instead of keeping the phone next to the bed, the user could place it directly into the SleepHelper charging dock.

The mobile application could record:

- ◆ Time at which the phone was placed in the dock,
- ◆ Estimated time spent using the phone before bedtime,
- ◆ Bedtime,
- ◆ Wake-up time,
- ◆ Compliance with the planned sleep routine.

The purpose is not to punish or restrict the user, but to provide feedback about behavioural patterns.

8. SleepScore

SleepHelper will include an experimental scoring system referred to as **SleepScore**.

The score will combine several categories of information, potentially including:

Sleep schedule

- ◆ planned bedtime,
- ◆ actual bedtime,
- ◆ wake-up time,
- ◆ regularity.

Sleep duration

- ◆ estimated duration,
 - ◆ deviation from the user's target.
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Environment

- ◆ temperature,
- ◆ humidity,
- ◆ light exposure,
- ◆ potentially CO₂ and noise.

Digital behaviour

- ◆ smartphone usage before bedtime,
- ◆ time at which the phone was placed in the charging dock.

Subjective assessment

The user will be able to rate their sleep quality after waking up.

For example:

SLEEPSCORE

87 / 100

Schedule 91

Duration 84

Environment 94

Phone 82

Self-rating 86

The weighting and methodology of the score will be developed experimentally and documented as part of the project.

SleepScore will not be presented as a medical measurement or diagnostic tool.

9. Mobile Application

A companion mobile application is planned as an extension of the physical device.

The application may provide:

- ◆ sleep history,
- ◆ SleepScore history,
- ◆ environmental data,
- ◆ bedtime consistency,
- ◆ smartphone behaviour,
- ◆ personalized recommendations,
- ◆ graphical representations of trends,
- ◆ configuration of alarms and sleep goals.

The long-term goal is for the application to identify patterns rather than simply display raw measurements.

For example:

Your average SleepScore was 12 points higher on days when you stopped using your phone at least 30 minutes before bedtime.

Such insights would be generated only after sufficient user data has been collected and analysed.

10. Physical Device

The physical SleepHelper unit is planned as a compact device designed for placement near the user's bed.

Two installation modes are considered:

Desktop configuration

- ◆ placed on a bedside table or other flat surface.

Wall-mounted configuration

- ◆ attached to a wall near the bed.

The device will incorporate:

- ◆ display,
- ◆ clock,
- ◆ controls,
- ◆ environmental sensors,
- ◆ charging dock,
- ◆ alarm,
- ◆ ambient lighting,
- ◆ power management.

The final mechanical design will be developed using CAD software.

11. Industrial Design

A significant part of the project will focus on the physical appearance and user experience of the device.

Several visual variants are planned.

Retro

- ◆ natural wood,
- ◆ cream-coloured components,
- ◆ warm display,
- ◆ physical buttons and/or rotary controls,
- ◆ subtle indicator lighting.

Minimal

- ◆ white or light grey body,
- ◆ simple geometry,
- ◆ restrained interface.

Industrial

- ◆ concrete-inspired materials,
- ◆ dark colours,
- ◆ metal elements,
- ◆ technical aesthetic.

Soft

- ◆ pastel colours,
 - ◆ light wood,
 - ◆ softer geometry.
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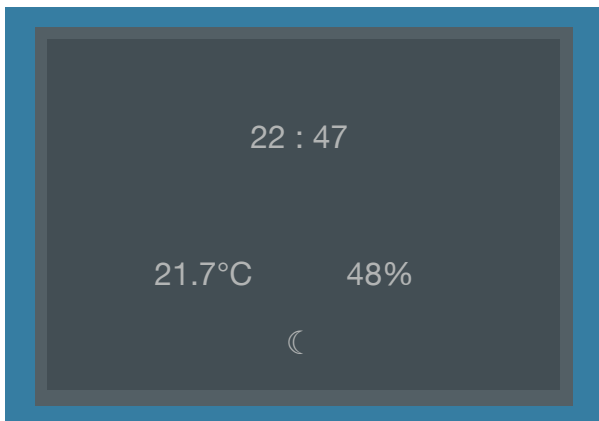
The different visual variants should ideally share the same internal electronic platform while allowing interchangeable or redesigned housings.

12. User Interface

The physical interface will be intentionally simple.

The device should provide information without becoming another source of distraction.

A possible main screen:



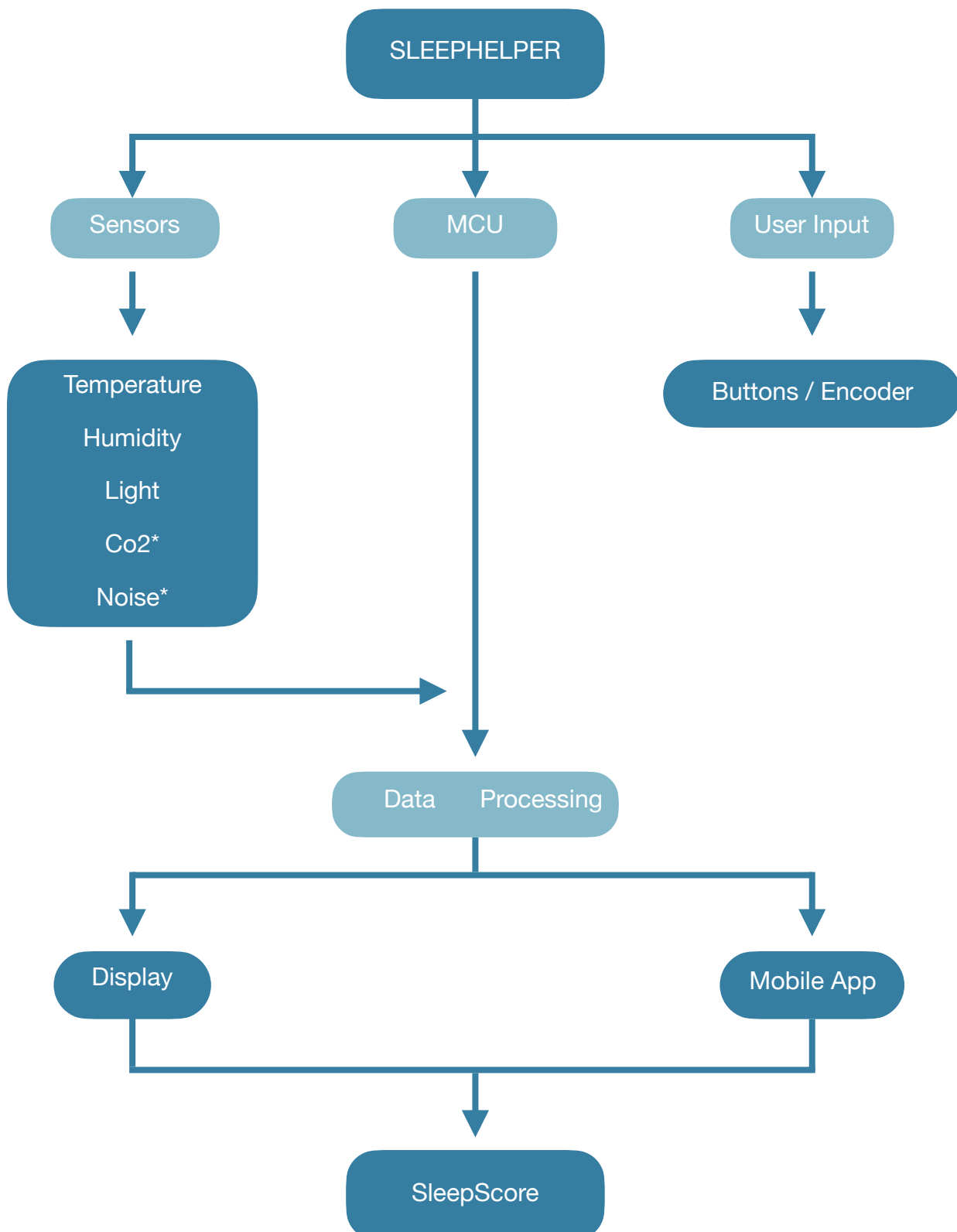
During the evening, the interface may automatically reduce its brightness.

The user should be able to configure:

- ◆ display brightness,
 - ◆ ambient light intensity,
 - ◆ alarm,
 - ◆ sleep schedule,
 - ◆ interface mode.
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13. Technical Architecture

The system will consist of several interconnected layers.



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* Future development.

The initial prototype will likely be based on a microcontroller platform such as Arduino-compatible hardware. The final hardware architecture will be determined during development.

14. Development Roadmap

Phase I — Research

- ◆ research on sleep hygiene,
- ◆ research on environmental factors,
- ◆ research on digital behaviour,
- ◆ analysis of existing solutions,
- ◆ identification of project requirements.

Phase II — Proof of Concept

- ◆ temperature measurement,
 - ◆ humidity measurement,
 - ◆ light measurement,
 - ◆ LCD interface,
 - ◆ clock,
 - ◆ alarm,
 - ◆ basic controls.
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Phase III — Embedded Prototype

- ◆ integrated sensors,
- ◆ RTC,
- ◆ power management,
- ◆ automatic brightness,
- ◆ user interaction,
- ◆ data storage.

Phase IV — Mobile Application

- ◆ device communication,
- ◆ data synchronization,
- ◆ visualization,
- ◆ user profiles,
- ◆ SleepScore prototype.

Phase V — Industrial Design

- ◆ CAD model,
 - ◆ enclosure prototypes,
 - ◆ wall mounting system,
 - ◆ desktop stand,
 - ◆ material and colour exploration.
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Phase VI — Testing

- ◆ hardware testing,
- ◆ software testing,
- ◆ sensor validation,
- ◆ usability testing,
- ◆ analysis of collected data.

Phase VII — Final Prototype

- ◆ custom PCB,
 - ◆ final enclosure,
 - ◆ integrated electronics,
 - ◆ mobile application,
 - ◆ complete documentation.
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15. Expected Outcome

The expected result of the project is a functional prototype of SleepHelper consisting of:

- ◆ Physical electronic device
- ◆ Embedded software
- ◆ Mobile application
- ◆ SleepScore analysis system
- ◆ Custom-designed enclosure
- ◆ Technical documentation
- ◆ Research documentation
- ◆ Prototype testing results

The final project should demonstrate the complete development process from problem identification and research to functional hardware and software implementation.

16. Future Development

Potential future developments include:

- ◆ improved sleep analysis,
 - ◆ additional environmental sensors,
 - ◆ more advanced personalization,
 - ◆ machine-learning-based pattern detection,
 - ◆ wireless communication,
 - ◆ custom PCB,
 - ◆ battery-powered operation,
 - ◆ wireless smartphone charging,
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- ◆ smart lighting,
- ◆ integration with other smart-home systems,
- ◆ additional enclosure designs.

The project may also serve as a foundation for future research into the relationship between environmental conditions, digital behaviour and subjective sleep quality.

17. Project Limitations

SleepHelper is intended as a ***wellbeing and habit-support system*** rather than a medical device.

The system will not attempt to diagnose sleep disorders or provide medical recommendations.

Any conclusions generated by the SleepScore system will be treated as experimental and personalized indicators rather than clinically validated measurements.

This distinction will remain an important part of the project's design and documentation.

18. References

This section will be expanded during the research phase.

The research will prioritize:

- ◆ peer-reviewed scientific publications,
 - ◆ systematic reviews,
 - ◆ publications from recognized medical and scientific organizations,
 - ◆ official public-health data,
 - ◆ technical documentation for hardware components.
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